张 晗 Han Zhang

副教授

清华大学 网络科学与网络空间研究院

中国 北京

zhhan@tsinghua.edu.cn • netsyscode.github.io/netsyslab • Google Scholar • github.com/netsyscode

研究概述

本人的研究面向 **AI / 网络基础设施**，构建**测量**、**资源调度与安全**技术，覆盖互联网（BGP、RPKI、DNS、Web PKI）、ISP 与运营商骨干网、**FITI / CERNET / CERNET2** 国家级试验设施，以及从智算集群到超大规模云与 Serverless 平台的智算 / 数据中心。研究工作分为三个方向，服务于同一个目标——让 *AI / 网络基础设施看得清、调得优、守得住*：

1. **AI / 网络基础设施测量**。以线速、零应用改造的方式让生产基础设施可观测：基于 eBPF 的零代码分布式追踪（SIGCOMM'23、SIGCOMM'25、ICSE'26）；SRv6 探测与时延瓶颈诊断（ICNP'25、ToN'26）；可编程交换机上的流测量（TPDS'22）；以及面向 Looking Glass 服务、AS 商业关系与域间路由的互联网级测量（WWW'26、ToN'25、ICNP'26）。
2. **AI / 网络基础设施资源调度**。把测量结果转化为高效、抗故障的带宽与算力调度：具备可用性保障的跨数据中心广域网流量工程（CoNEXT'21、ToN'22）；集中式网络效用最大化（ICNP'22、ToN'25）；路由代数（INFOCOM'25）；可编程交换机上的网内加速（INFOCOM'24、TPDS'22）；面向网络功能的 FaaS 平台（TPDS'23）；服务功能链部署与调度（TPDS'26）；以及面向企业云的即插即用网络管理（NSDI'27）。
3. **AI / 网络基础设施安全**。在攻击与配置错误扩散之前发现并阻断：可编程交换机上的在线入侵检测，已在 ISP 生产环境运行（SIGCOMM'25、SIGCOMM'26）；可解释、能应对正常态漂移的异常检测（CCS'21、NDSS'23）；基于溯源图的 APT 检测（TIFS'22、TIFS'25）；Serverless 钱包耗尽（DoW）攻击检测（CCS'22、ToN'26）；路由与 Web PKI 安全（CCS'26、WWW'25）；以及运营商规模的配置验证（TIFS'24、CoNEXT'25、INFOCOM'26）。

学术计量

1,730	17	29	近 100	13
引用次数	h 指数	i10 指数	论文	在读学生

数据来源 Google Scholar，2026 年。已在 ACM SIGCOMM、USENIX NSDI、ACM CCS、NDSS、ACM CoNEXT、ACM WWW、ACM ICSE、IEEE INFOCOM、IEEE/ACM ToN、IEEE TIFS、IEEE TPDS 等国际顶级会议与期刊发表论文近百篇。

教育背景

2013.09 – 2018.07	**工学博士**，计算机科学与技术 清华大学，中国 北京　导师：尹霞教授
2009.09 – 2013.07	**工学学士**，计算机科学与技术 吉林大学，中国 长春

工作履历

现任	**副教授**、博士生导师 清华大学网络科学与网络空间研究院
曾任	**助理教授** 清华大学网络科学与网络空间研究院

主要奖励与荣誉

- **Applied Networking Research Prize（ANRP）**，IRTF / Internet Society，2023。国际

- **中国通信学会科学技术奖一等奖**——《城域云网关键技术及规模应用》，2025。国家级学会
- **NSGA 杰出研究者奖（Best Researcher Award）**，2025。
- **ACM TURC SIGCSE 中国新星奖**，2023。
- **优秀服务奖（Excellent Service Award）**，IEEE/ACM ICA3PP 2023。

代表性论文

加粗为本人；* 表示通信作者。完整列表见下一节。

AI / 网络基础设施测量

[1] Junxian Shen, **Han Zhang***, Yang Xiang, Xingang Shi, Xinrui Li, Yunxi Shen, Zijian Zhang, Sicheng Wang, Yanhui Yang, Yu Zhou, Mingwei Xu, Jilong Wang. "Network-Centric Distributed Tracing with DeepFlow: Troubleshooting Your Microservices in Zero Code." *ACM SIGCOMM 2023*, pp. 420–437.
↪ 以网络为中心的追踪平面与隐式上下文传播，完全无需修改应用代码。已开源并成为云原生可观测性的事实标准（CNCF Sandbox，4000+ stars，部署于数万台主机）。

[2] Yantao Geng, **Han Zhang***, Zhiheng Wu, Yahui Li, Jilong Wang, Xia Yin. "Low-Overhead Distributed Application Observation with DeepTrace: Achieving Accurate Tracing in Production Systems." *ACM SIGCOMM 2025*, pp. 1056–1069.
↪ 基于事务的无侵入追踪：利用请求内容中的 API 端点与事务字段将请求归入不同事务，避免误合并不同事务的追踪；高并发下准确率仍超过 95%，已被数十家企业用于生产环境。

[3] Yantao Geng, **Han Zhang***, Zhiheng Wu, Yahui Li, Jilong Wang, Xia Yin. "Non-Intrusive Distributed Tracing with Method-Level Delay Estimation for Microservices Troubleshooting." *ACM/IEEE ICSE 2026*, pp. 1661–1673.
↪ FlowScope 通过互相关分析估计请求间的方法级因果时延分布，并在多重约束下迭代重建追踪，追踪准确率相比 DeepFlow 与 TraceWeaver 提升 51%。

[4] Kaiyang Zhao, **Han Zhang***, Yao Tong, Yahui Li, Xingang Shi, Zhiliang Wang, Xia Yin, Jianping Wu. "SRmesh: Deterministic and Efficient Diagnosis of Latency Bottleneck Links in SRv6 Networks." *IEEE ICNP 2025*, pp. 1–12.
↪ 以确定性路径的 SR 探测、基于时延的多路径推断与渐进式诊断，定位 SRv6 网络中的时延瓶颈链路，探测开销最多降低 91.9%。

[5] Yunze Wei, Xingang Shi, **Han Zhang***, Tianyu Zhang, Yahui Li, Xia Yin. "GlassMiner: Mining Looking Glass Services via Structure-Semantics Fusion for Web Observability." *ACM WWW 2026*, pp. 7576–7587.
↪ 融合结构与语义特征并借助大模型分类，发现 3867 个 Looking Glass 网页和 7558 个此前未被公开列表收录的观测点（发现量为已有方法的 1.82 倍、挖掘效率为 3.43 倍），并据此分析两家大型 CDN 的路由绕行。

AI / 网络基础设施资源调度

[6] Kaiyang Zhao, **Han Zhang***, Tianpeng Zhang, Yantu Tong, Zeji Xiao, Xuefeng Liu, Xia Yin, Jianping Wu. "Kuiper: Enterprise Cloud Plug-and-Play Network Management Platform." *USENIX NSDI 2027*.
↪ 面向企业云的统一智能运维控制面：在同一套遥测与验证基底上整合故障管理、安全事件响应与配置管理。

[7] **Han Zhang**, Xingang Shi*, Xia Yin, Jilong Wang, Zhiliang Wang, Yingya Guo, Tian Lan. "Boosting Bandwidth Availability over Inter-DC WAN." *ACM CoNEXT 2021*, pp. 297–312.
↪ 面向跨数据中心广域网的带宽可用性流量工程：在线准入控制比最优策略快 30×（误拒率低于 4%），相比 FFC 与 TEAVAR 多满足 23%–60% 的可用性需求。

[8] Yi Qiao, **Han Zhang***, Jilong Wang. "NetFEC: In-Network FEC Encoding Acceleration for Latency-Sensitive Multimedia Applications." *IEEE INFOCOM 2024*, pp. 2348–2357.
↪ 将时延敏感多媒体应用的 FEC 编码整体卸载到可编程交换芯片，以 ghost packet 机制兼容拥塞控制、

发送节奏控制与统计功能；与 WebRTC 集成后消除服务器 CPU 负担，额外开销可忽略。

[9] Heng Yu, **Han Zhang***, Junxian Shen, Yantao Geng, et al. "Serpens: A High-Performance FaaS Platform for Network Functions." *IEEE Transactions on Parallel and Distributed Systems*, vol. 34, no. 8, pp. 2448–2463, 2023.

↪ 面向网络功能的高性能 FaaS 平台：可复用的 NF 运行时、本地状态访问与改进的服务链机制，相比在现有 FaaS 平台上运行的 NF 与服务链，吞吐提升 10 倍以上、时延降低 90% 以上。

[10] Ying Tian, Zhiliang Wang*, Xia Yin, Xingang Shi, Jiahai Yang, **Han Zhang***. "Centralized Network Utility Maximization with Accelerated Gradient Method." *IEEE Transactions on Networking*, 2025.

↪ 通过光滑的效用与惩罚函数，使 Nesterov 加速梯度法可用于集中式网络效用最大化，收敛速率达 $O(d/t^2)$，对流数量可扩展，并支持多路径上的带优先级流。

AI / 网络基础设施安全

[11] **Han Zhang**, Xuefeng Liu, Linqiang Qian, Guyue Liu, Tianyu Zhang, Kaiyang Zhao, Yantu Tong, Zeji Xiao, Dongbiao He, Yahui Li, Ke Ruan, Jilong Wang, Yongqing Zhu, Xia Yin. "Towards High-Performance Intrusion Detection with Robustness Guarantees on Programmable Switches at ISP Scale." *ACM SIGCOMM 2026*, pp. 1282–1298.

↪ 部署于可编程交换机的在线入侵检测系统，保护接入 ISP 的企业园区：双平面特征提取、仅用良性流量训练的单类模型与在线更新机制；已在生产环境运行三年以上，可在数分钟内以约 90% 的准确率检出恶意流量。

[12] **Han Zhang**, Guyue Liu*, Xingang Shi, Yahui Li, Jilong Wang, Yongqing Zhu, Ke Ruan, Jie Liang, Xia Yin. "Achieving High-Speed and Robust Encrypted Traffic Anomaly Detection with Programmable Switches." *ACM SIGCOMM 2025*, pp. 1254–1256.

↪ 双平面特征提取 + 轻量单类分类模型，化解深度学习精度与交换芯片资源约束之间的长期矛盾；已在生产环境运行两年，发现 10 余起重大攻击事件，为两家 ISP 保护超过 1000 万台服务器。

[13] Yunze Wei, Xingang Shi, Minghao Zhang, Zhaozhen Wang, Qiang Ma, **Han Zhang**, Yahui Li, Xia Yin. "Parade: Unified and Data-Driven Prefix-AS Semantics for BGP Anomaly Detection." *ACM CCS 2026*.

↪ 统一的数据驱动前缀—AS 语义模型，把碎片化的 BGP 观测转化为可用于异常检测的一致性基准事实。

[14] Junxian Shen, **Han Zhang***, Yantao Geng, Jiawei Li, Jilong Wang, Mingwei Xu. "Gringotts: Fast and Accurate Internal Denial-of-Wallet Detection for Serverless Computing." *ACM CCS 2022*, pp. 2627–2641.

↪ 在商用 Serverless 平台上实测内部钱包耗尽（DoW）攻击，并以指标采集代理结合马氏距离异常分析进行检测：性能开销低于 1.1%，平均检测时延 1.86 秒，准确率超过 95.75%。

[15] Dongqi Han, Zhiliang Wang*, Wenqi Chen, Ying Zhong, Su Wang, **Han Zhang**, Jiahai Yang, Xingang Shi, Xia Yin. "DeepAID: Interpreting and Improving Deep Learning-based Anomaly Detection in Security Applications." *ACM CCS 2021*, pp. 3197–3217.

↪ 面向安全领域无监督深度异常检测模型的解释框架，使运维人员能够诊断模型误判并显著降低误报率。

[16] Dongqi Han, Zhiliang Wang*, Wenqi Chen, Kai Wang, Rui Yu, Su Wang, **Han Zhang**, et al. "Anomaly Detection in the Open World: Normality Shift Detection, Explanation, and Adaptation." *NDSS 2023*.

↪ 针对零正样本异常检测中的正常态漂移（仅用良性数据训练的模型在正常行为变化时受影响最大），实现漂移的检测、解释与自适应。

[17] Tianyu Zhang, **Han Zhang***, Yunze Wei, Yahui Li, Xingang Shi, Jilong Wang, Xia Yin. "ACME++: A Secure Authorization Mechanism for ACME Clients in the Web PKI Ecosystem." *ACM WWW 2025*, pp. 1058–1067.

↪ 披露 *ACME Authz 缓存攻击*——攻击者无需控制域名即可获取欺诈证书（在 Let's Encrypt 上实证），并提出开销可忽略的账户绑定缓解机制。

[18] XinZhe Liu, Yahui Li*, **Han Zhang**, Xia Yin, Xingang Shi, Zhiliang Wang, Gang Ren, Jilong Wang, Jiangyuan Yao. "Scalable and Interpretable Overlay Network Checking via Ensemble

Verification." *ACM CoNEXT 2025*.
↪ 首个面向 overlay/underlay 网络的控制平面验证器；部署于全国性 FITI 试验设施，发现 22 处配置错误，相比 Batfish+ 最高提速 102×。

[19] Yahui Li, **Han Zhang***, Jilong Wang, Xingang Shi, Xia Yin, Zhiliang Wang, Jiankun Hu, Congcong Miao, Jianping Wu. "Proactively Verifying Quantitative Network Policy across Unsafe and Unreliable Environments." *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 10099–10113, 2024.
↪ 定量网络验证（QNV）不仅检查可达性等布尔策略，还能检查链路过载等定量策略，并覆盖故障场景；借助收敛转发的符号化模型，大规模网络也能在数分钟内完成验证。

论文列表（按年份）

加粗为本人；* 表示通信作者。

2027

[1] Kaiyang Zhao, **Han Zhang***, Tianpeng Zhang, Yantu Tong, Zeji Xiao, Xuefeng Liu, Xia Yin, Jianping Wu. "Kuiper: Enterprise Cloud Plug-and-Play Network Management Platform." *USENIX NSDI 2027*. （待刊）

2026

[1] **Han Zhang**, Xuefeng Liu, Linqiang Qian, Guyue Liu, Tianyu Zhang, Kaiyang Zhao, Yantu Tong, Zeji Xiao, Dongbiao He, Yahui Li, Ke Ruan, Jilong Wang, Yongqing Zhu, Xia Yin. "Towards High-Performance Intrusion Detection with Robustness Guarantees on Programmable Switches at ISP Scale." *ACM SIGCOMM 2026*, pp. 1282–1298.

[2] Yunze Wei, Xingang Shi, Minghao Zhang, Zhaozhen Wang, Qiang Ma, **Han Zhang**, Yahui Li, Xia Yin. "Parade: Unified and Data-Driven Prefix-AS Semantics for BGP Anomaly Detection." *ACM CCS 2026*. （待刊）

[3] Yunze Wei, Xingang Shi, **Han Zhang***, Tianyu Zhang, Yahui Li, Xia Yin. "GlassMiner: Mining Looking Glass Services via Structure-Semantics Fusion for Web Observability." *ACM WWW 2026*, pp. 7576–7587.

[4] Yantao Geng, **Han Zhang***, Zhiheng Wu, Yahui Li, Jilong Wang, Xia Yin. "Non-Intrusive Distributed Tracing with Method-Level Delay Estimation for Microservices Troubleshooting." *ACM/IEEE ICSE 2026*, pp. 1661–1673.

[5] Junxian Shen, **Han Zhang***, Wenxue Lin, Yantao Geng, Jiawei Li, Jilong Wang, Mingwei Xu. "Clover: Workload Verification for Real-Time Detection of Contention-Induced Slowdowns in Serverless Platforms." *IEEE/ACM Transactions on Networking*, vol. 34, pp. 4700–4715, 2026.

[6] Kaiyang Zhao, **Han Zhang***, Yao Tong, Yahui Li, Xingang Shi, Zhiliang Wang, Xia Yin, Jianping Wu. "Efficient Slice-Parallel Distributed Probing in SRv6 Networks." *IEEE/ACM Transactions on Networking*, vol. 34, pp. 5588–5600, 2026.

[7] XinZhe Liu, Yahui Li, **Han Zhang**, Renrui Tian, Xia Yin, Xingang Shi, Zhiliang Wang, Gang Ren, Jilong Wang, Jiangyuan Yao. "Comprehensive Network Configuration Verification via Effective Environment Reduction." *IEEE INFOCOM 2026*, pp. 1–10.

[8] Letong Sun, Xingang Shi, Fengyan Han, Liyuan Chang, Xia Yin, Zhiliang Wang, **Han Zhang**. "A Comparative Approach for Studying Control-Plane and Data-Plane Mismatch in Interdomain Routing." *IEEE ICNP 2026*. （待刊）

[9] Letong Sun, Xingang Shi, Fengyan Han, Xia Yin, Zhiliang Wang, **Han Zhang**. "The Anatomy of ROV Rollout: A Sub-AS and Longitudinal Perspective." *IEEE ICNP 2026*. （待刊）

[10] Weihan Chen, Zhiliang Wang, **Han Zhang**, Xia Yin, Xingang Shi. "Delay-Aware Service Function Chain Embedding and Scheduling in Edge Clouds." *IEEE Transactions on Parallel and Distributed Systems*, vol. 37, no. 11, pp. 2539–2553, 2026.

[11] Shuhuan Wang, Zhiyu Wang, Hao Sun, Zhi Jiang, Xia Yin, Dongbin Han, **Han Zhang**, Xingang Shi, Jiahai Yang. "The Evolution of Provenance-Based Intrusion Detection Systems: A Review of Approaches, Limitations, and Future Directions." *IEEE Network*, 2026.

[12] 李秋香, **张晗***, 曲豫宾, 吴建平. "MASTER：基于知识注入与意图强化的大语言模型漏洞验证代码生成框架." 电子学报, vol. 54, no. 4, pp. 1802–1822, 2026.

2025

[1] Tianyu Zhang, **Han Zhang***, Yunze Wei, Yahui Li, Xingang Shi, Jilong Wang, Xia Yin. "ACME++: A Secure Authorization Mechanism for ACME Clients in the Web PKI Ecosystem." *ACM WWW 2025*, pp. 1058–1067.

[2] Yantao Geng, **Han Zhang***, Zhiheng Wu, Yahui Li, Jilong Wang, Xia Yin. "Low-Overhead Distributed Application Observation with DeepTrace: Achieving Accurate Tracing in Production Systems." *ACM SIGCOMM 2025*, pp. 1056–1069.

[3] **Han Zhang**, Guyue Liu*, Xingang Shi, Yahui Li, Jilong Wang, Yongqing Zhu, Ke Ruan, Jie Liang, Xia Yin. "Achieving High-Speed and Robust Encrypted Traffic Anomaly Detection with Programmable Switches." *ACM SIGCOMM 2025*, pp. 1254–1256.

[4] XinZhe Liu, Yahui Li*, **Han Zhang**, Xia Yin, Xingang Shi, Zhiliang Wang, Gang Ren, Jilong Wang, Jiangyuan Yao. "Scalable and Interpretable Overlay Network Checking via Ensemble Verification." *ACM CoNEXT 2025*.

[5] Renrui Tian, Yahui Li*, **Han Zhang**, XinZhe Liu, Xia Yin, Xingang Shi, Zhiliang Wang, Jilong Wang. "Poster: ERIS — Evaluating ROV via ICMPv6 Rate Limiting Side Channels." *ACM CCS 2025*.

[6] Kaiyang Zhao, **Han Zhang***, Yao Tong, Yahui Li, Xingang Shi, Zhiliang Wang, Xia Yin, Jianping Wu. "SRmesh: Deterministic and Efficient Diagnosis of Latency Bottleneck Links in SRv6 Networks." *IEEE ICNP 2025*, pp. 1–12.

[7] Zhaozhen Wang, Xingang Shi*, Haijun Geng, **Han Zhang**, Xia Yin, Zhiliang Wang. "On Non-Commutative Routing." *IEEE INFOCOM 2025*, pp. 1–10.

[8] Zhiyun Tang, Yahui Li, Xingang Shi, Jilong Wang, Yongqing Zhu, Ke Ruan, Xia Yin, **Han Zhang***. "Probabilistic Analysis of Overload-Free Property for Critical Traffic." *IEEE/ACM IWQoS 2025*, pp. 1–6.

[9] Su Wang, Hongbin Sun, Zhiliang Wang, Tao Zhou, Xia Yin, Dongqi Han, **Han Zhang**, Xingang Shi, Jiahai Yang. "End-to-End Attack Scene Reconstruction in a Host with Rules and Anomaly-Based Detection Models." *IEEE Transactions on Information Forensics and Security*, 2025.

[10] Ying Tian, Zhiliang Wang*, Xia Yin, Xingang Shi, Jiahai Yang, **Han Zhang***. "Centralized Network Utility Maximization with Accelerated Gradient Method." *IEEE Transactions on Networking*, 2025.

[11] Xingang Shi, Zitong Jin*, Bin Xiong, Xinyao Huang, Xiaotian Xi, Dan Li, **Han Zhang**, Zhiliang Wang, Xia Yin. "HELA: Inferring AS Relationships with a Hybrid of Empirical and Learning Algorithms." *IEEE Transactions on Networking*, 2025.

[12] Weiting Zhang, Jiadong Ren, Tao Zheng, Ruibin Guo, **Han Zhang**, Shiwen Mao, Hongke Zhang*. "GenNet: Computing-Efficient Generative AI for Deterministic Transmission Scheduling in 6G Networks." *IEEE Communications Magazine*, vol. 63, no. 8, pp. 48–55, 2025.

[13] Yanxu Fu, Pei Zhang*, **Han Zhang**, Xiaohong Huang, Kun Xie, Dandan Li. "BGPFlow: Flow-Based Feature Extraction for BGP Anomaly Detection." *ACM SIGCOMM NGNO Workshop 2025*, pp. 13–20.

2024

[1] Yi Qiao, **Han Zhang***, Jilong Wang. "NetFEC: In-Network FEC Encoding Acceleration for Latency-Sensitive Multimedia Applications." *IEEE INFOCOM 2024*, pp. 2348–2357.

[2] Yahui Li, **Han Zhang***, Jilong Wang, Xingang Shi, Xia Yin, Zhiliang Wang, Jiankun Hu, Congcong Miao, Jianping Wu. "Proactively Verifying Quantitative Network Policy across Unsafe and Unreliable Environments." *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 10099–10113, 2024.

[3] XinZhe Liu, Yahui Li*, **Han Zhang**, Xia Yin, Xingang Shi, Zhiliang Wang, Gang Ren, Jiangyuan Yao. "Poster: Scalable and Interpretable Multilayer Overlay Network Checking via Ensemble Verification." *ACM SIGCOMM 2024 Posters & Demos*.

[4] Weihan Chen, Zhiliang Wang*, **Han Zhang**, Xia Yin, Xingang Shi. "Cost-Efficient Flow Migration for SFC Dynamical Scheduling in Geo-Distributed Clouds." *Computer Networks*, vol. 249, 2024.

[5] Jun He, Yahui Li*, **Han Zhang***, Changqing An, Jilong Wang. "ROV-GD: Improving the Measurement of ROV Deployment Using Graph Difference." *IEEE ISCC 2024*, pp. 1–7.

[6] Kunling He, Fenghua Li, Jessie Hui Wang, **Han Zhang**, Yiren Zhao. "AB-TCAD: An Access Behavior-Based Two-Stage Compromised Account Detection Framework." *IFIP Networking 2024*, pp. 104–112.

2023

[1] Junxian Shen, **Han Zhang***, Yang Xiang, Xingang Shi, Xinrui Li, Yunxi Shen, Zijian Zhang, et al. "Network-Centric Distributed Tracing with DeepFlow: Troubleshooting Your Microservices in Zero Code." *ACM SIGCOMM 2023*, pp. 420–437.

[2] Dongqi Han, Zhiliang Wang*, Wenqi Chen, Kai Wang, Rui Yu, Su Wang, **Han Zhang**, et al. "Anomaly Detection in the Open World: Normality Shift Detection, Explanation, and Adaptation." *NDSS 2023*.

[3] Heng Yu, **Han Zhang***, Junxian Shen, Yantao Geng, et al. "Serpens: A High-Performance FaaS Platform for Network Functions." *IEEE Transactions on Parallel and Distributed Systems*, vol. 34, no. 8, pp. 2448–2463, 2023.

[4] Yahui Li, **Han Zhang***, Jilong Wang, Xia Yin, Xingang Shi, Jianping Wu. "A General Approach to Generate Test Packets with Network Configurations." *IEEE Transactions on Parallel and Distributed Systems*, vol. 34, no. 4, pp. 1362–1375, 2023.

[5] Pei Zhang, Fangzhou He, **Han Zhang***, Jiankun Hu, et al. "Real-Time Malicious Traffic Detection with Online Isolation Forest over SD-WAN." *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 2076–2090, 2023.

[6] Tianbo Wang, Huacheng Li, Chunhe Xia*, **Han Zhang**, Pei Zhang. "From the Dialectical Perspective: Modeling and Exploiting of Hybrid Worm Propagation." *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 1610–1624, 2023.

[7] Yantao Geng, **Han Zhang***, Jilong Wang. "Delay-Based Congestion Control for Cross-Datacenter Networks." *IEEE/ACM IWQoS 2023*.

[8] Jiawei Li, **Han Zhang***, Jilong Wang. "PUVAR: Minimize Idle Resource SLO Violations by Uncertainty-Aware Scheduling in Cloud Platforms." *IEEE ICWS 2023*.

[9] Ming Wang, Yahui Li, **Han Zhang**, Anlun Hong, Jun He, Jilong Wang. "Knocking Cells: Latency-Based Identification of IPv6 Cellular Addresses on the Internet." *IEEE ISCC 2023*.

[10] Allen Hong, Yahui Li*, **Han Zhang***, Ming Wang, Changqing An, Jilong Wang. "A Cheap and Accurate Delay-Based IP Geolocation Method Using Machine Learning and Looking Glass." *IFIP Networking 2023*.

[11] Jun He, Yahui Li*, **Han Zhang***, Ming Wang, Changqing An, Jilong Wang. "Be Careful of Your Neighbors: Injected Sub-Prefix Hijacking Invisible to Public Monitors." *IEEE ICC 2023*.

[12] Shuanghong Yu, Fenghua Li*, **Han Zhang**, Kunling He, Xingkun Yao. "GapReplay: A High-Accuracy Packet Replayer." *IEEE ICC 2023*.

2022

[1] Junxian Shen, **Han Zhang***, Yantao Geng, Jiawei Li, Jilong Wang, Mingwei Xu. "Gringotts: Fast and Accurate Internal Denial-of-Wallet Detection for Serverless Computing." *ACM CCS 2022*, pp. 2627–2641.

[2] **Han Zhang**, Xia Yin, Xingang Shi*, Jilong Wang, Zhiliang Wang, Yingya Guo, Tian Lan, Yahui Li, Yongqing Zhu, Ke Ruan, Haijun Geng. "Achieving High Availability in Inter-DC WAN Traffic Engineering." *IEEE/ACM Transactions on Networking*, 2022.

[3] Su Wang, Zhiliang Wang*, Tao Zhou, Hongbin Sun, Xia Yin, Dongqi Han, **Han Zhang**, et al. "THREATRACE: Detecting and Tracing Host-Based Threats in Node Level through Provenance Graph Learning." *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 3972–3987, 2022.

[4] Yi Qiao, Menghao Zhang, Yu Zhou, Xiao Kong, **Han Zhang***, et al. "NetEC: Accelerating Erasure Coding Reconstruction with In-Network Aggregation." *IEEE Transactions on Parallel and Distributed Systems*, vol. 33, no. 10, pp. 2571–2583, 2022.

[5] Zongyi Zhao, Xingang Shi*, Zhiliang Wang, Qi Li, **Han Zhang**, Xia Yin. "Efficient and Accurate Flow Record Collection with HashFlow." *IEEE Transactions on Parallel and Distributed Systems*, vol. 33, no. 5, pp. 1069–1083, 2022.

[6] Ying Tian, Zhiliang Wang*, Xia Yin, Xingang Shi, Jiahai Yang, **Han Zhang**. "Centralized Network Utility Maximization with Accelerated Gradient Method." *IEEE ICNP 2022*.

[7] Heng Yu, Junxian Shen, **Han Zhang***, Jilong Wang, Congcong Miao, Mingwei Xu. "Scorpius: Proactive Code Preparation to Accelerate Function Startup." *IEEE/ACM IWQoS 2022*.

[8] Shangbin Han, Qianhong Wu, **Han Zhang***, Bo Qin. "Light-Weight Unsupervised Anomaly Detection for Encrypted Malware Traffic." *IEEE DSC 2022*.

[9] Shangbin Han, Qianhong Wu, **Han Zhang**, Bo Qin, Jiangyuan Yao, Willy Susilo. "Autoperman: Automatic Network Traffic Anomaly Detection with Ensemble Learning." *ICAIS 2022*.

[10] Yongkang Chen, Ming Zhang*, Jin Li, Xiaohui Kuang, Xurong Zhang, **Han Zhang**. "Dynamic and Diverse Transformations for Defending against Adversarial Examples." *IEEE TrustCom 2022*.

2021

[1] **Han Zhang**, Xingang Shi*, Xia Yin, Jilong Wang, Zhiliang Wang, Yingya Guo, Tian Lan. "Boosting Bandwidth Availability over Inter-DC WAN." *ACM CoNEXT 2021*, pp. 297–312.

[2] Dongqi Han, Zhiliang Wang*, Wenqi Chen, Ying Zhong, Su Wang, **Han Zhang**, Jiahai Yang, Xingang Shi, Xia Yin. "DeepAID: Interpreting and Improving Deep Learning-Based Anomaly Detection in Security Applications." *ACM CCS 2021*, pp. 3197–3217.

[3] Shangbin Han, Qianhong Wu, **Han Zhang***, Bo Qin, Jiankun Hu, et al. "Log-Based Anomaly Detection with Robust Feature Extraction and Online Learning." *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 2300–2311, 2021.

[4] Yingya Guo, Weipeng Wang, **Han Zhang**, Wenzhong Guo, Zhiliang Wang, Ying Tian, et al. "Traffic Engineering in Hybrid Software-Defined Network via Reinforcement Learning." *Journal of Network and Computer Applications*, vol. 189, 2021.

[5] Weihan Chen, Zhiliang Wang*, **Han Zhang**, Xia Yin, Xingang Shi. "Cost-Efficient Dynamic Service Function Chain Embedding in Edge Clouds." *IEEE CNSM 2021*.

[6] Ying Tian, Zhiliang Wang*, Xia Yin, Xingang Shi, Jiahai Yang, **Han Zhang**, Yingya Guo, et al. "Traffic Engineering with Segment Routing Considering Probabilistic Failures." *IEEE CNSM 2021*.

[7] Haijun Geng, **Han Zhang***, Yangyang Zhang. "Efficient Routing Protection Algorithm in Large-Scale Networks." *Computers, Materials & Continua*, vol. 66, no. 2, pp. 1733–1744, 2021.

2020

[1] Yahui Li, Zhiliang Wang*, Xia Yin, Xingang Shi, Jianping Wu, Fangdan Ye, Jiangyuan Yao, **Han Zhang**. "Assisting Reachability Verification of Network Configurations Updates with NUV." *Computer Networks*, vol. 177, 2020.

[2] Haijun Geng, **Han Zhang**, Xingang Shi*, Zhiliang Wang, Xia Yin. "Efficient Computation of Loop-Free Alternates." *Journal of Network and Computer Applications*, vol. 151, 2020.

[3] Haijun Geng, **Han Zhang**, Xingang Shi*, et al. "A Hybrid Link Protection Scheme for Ensuring Network Service Availability in Link-State Routing Networks." *Journal of Communications and Networks*, vol. 22, no. 1, pp. 46–60, 2020.

[4] Weihan Chen, Zhiliang Wang, **Han Zhang**, Xia Yin, Xingang Shi, Lei Sun. "Joint Optimization of Service Function Chain Elastic Scaling and Routing." *IEEE INFOCOM Workshops 2020*.

2019

[1] **Han Zhang**, Haijun Geng, Yahui Li, et al. "DA&FD: Deadline-Aware and Flow Duration-Based Rate Control for Mixed Flows in DCNs." *IEEE/ACM Transactions on Networking*, vol. 27, no. 6, pp. 2458–2471, 2019.

[2] Zhiliang Wang, **Han Zhang***, Xingang Shi, et al. "Efficient Scheduling of Weighted Coflows in Data Centers." *IEEE Transactions on Parallel and Distributed Systems*, vol. 30, no. 9, pp. 2003–2017, 2019.

[3] Yahui Li, Xia Yin, Zhiliang Wang*, Jiangyuan Yao, Xingang Shi, Jianping Wu, **Han Zhang**, et al. "A Survey on Network Verification and Testing with Formal Methods: Approaches and Challenges." *IEEE Communications Surveys & Tutorials*, vol. 21, no. 1, pp. 940–969, 2019.

[4] Yahui Li, Zhiliang Wang*, Jiangyuan Yao, Xia Yin, Xingang Shi, Jianping Wu, **Han Zhang**. "MSAID: Automated Detection of Interference in Multiple SDN Applications." *Computer Networks*, vol. 153, 2019.

[5] Yingya Guo, Zhiliang Wang*, **Han Zhang**, Xia Yin, Xingang Shi, Jianping Wu. "Joint Optimization of Tasks Placement and Routing to Minimize Coflow Completion Time." *Journal of Network and Computer Applications*, vol. 135, 2019.

2018 年及以前

[1] Xiao Ma, Chuang Lin, **Han Zhang***, Jianwei Liu. "Energy-Aware Computation Offloading of IoT Sensors in Cloudlet-Based Mobile Edge Computing." *Sensors*, 2018.

[2] Zhifeng Liu, Xia Yin, Zhiliang Wang, Xingang Shi, Yiqiang Hua, Xiaohong Huang, Liang Wei, Jiangyuan Yao, **Han Zhang**, et al. "Design and Construction of Hierarchical and Cross-Regional SDN Verification and Demonstration System." *Telecommunications Science*, 2018.

[3] **Han Zhang**, Xingang Shi*, Xia Yin, Zhiliang Wang. "Yosemite: Efficient Scheduling of Weighted Coflows in Data Centers." *IEEE ICNP 2017*.

[4] **Han Zhang**, Xingang Shi*, Yingya Guo, Zhiliang Wang, Xia Yin. "More Load, More Differentiation — Let More Flows Finish before Deadline in Data Center Networks." *Computer Networks*, vol. 127, 2017.

[5] **Han Zhang**, Xingang Shi, Yingya Guo, Haijun Geng, Zhiliang Wang, Xia Yin. "Joint Source Selection and Transfer Optimization for Erasure Coding Storage System." *IEEE IPCCC 2017*.

[6] **Han Zhang**, Xingang Shi, Xia Yin, Zhiliang Wang, Yingya Guo. "FDRC: Flow Duration Time

Based Rate Control in Data Center Networks." *IEEE/ACM IWQoS 2016*.

[7] M. Gharaibeh, **Han Zhang**, C. Papadopoulos, J. Heidemann. "Assessing Co-Locality of IP Blocks." *IEEE INFOCOM Workshops 2016*.

[8] Zongyi Zhao, Qi Li, Mingwei Xu, Xingang Shi, **Han Zhang**. "Reduce Completion Time and Guarantee Throughput by Transport with Slight Congestion." *IEEE ICC 2016*.

[9] **Han Zhang**, Xingang Shi, Xia Yin, Fengyuan Ren, Zhiliang Wang. "More Load, More Differentiation — A Design Principle for Deadline-Aware Congestion Control." *IEEE INFOCOM 2015*.

[10] Haijun Geng, Xingang Shi, Xia Yin, Zhiliang Wang, **Han Zhang**. "Algebra and Algorithms for Efficient and Correct Multipath QoS Routing in Link State Networks." *IEEE/ACM IWQoS 2015*.

[11] Haijun Geng, Xingang Shi, Xia Yin, Zhiliang Wang, **Han Zhang**. "An Efficient Link Protection Scheme for Link-State Routing Networks." *IEEE ICC 2015*.

[12] Yingya Guo, Zhiliang Wang, Xia Yin, Xingang Shi, Jianping Wu, **Han Zhang**. "Incremental Deployment for Traffic Engineering in Hybrid SDN Network." *IEEE IPCCC 2015*.

[13] **Han Zhang**, C. Papadopoulos. "BotTalker: Generating Encrypted, Customizable C&C Traces." *IEEE HST 2015*.

[14] Haijun Geng, Xingang Shi, Xia Yin, Zhiliang Wang, **Han Zhang**, Jiangyuan Yao. "A Hybrid Link Protection Scheme for Link-State Routing Networks." *IEEE IPCCC 2014*.

[15] Haijun Geng, Xingang Shi, Xia Yin, Zhiliang Wang, **Han Zhang**, Jiangyuan Yao. "Let More Nodes Have a Second Choice." *IEEE IPCCC 2014*.

科研项目

- **国家自然科学基金面上项目**（No. 62572269）——基于网络行为分析的分布式微服务架构应用异常检测技术研究。**项目负责人**。
- **国家自然科学基金青年项目**（No. 62002009）——面向批量传输的数据中心广域网资源分配研究。**项目负责人**。
- **国家自然科学基金重大项目**（No. 62394322）——智算融合网络组网与路由关键技术研究。**子课题负责人**。
- **国家自然科学基金重点项目**（No. 61932011）——基于区块链的物联网安全技术研究。**子课题负责人**。
- **工信部重点研发项目**（No. TC220A04A-78）——高精度北斗车规级定位芯片及终端定位系统。**项目负责人**。
- **科技部国家重点研发计划**（No. 2018YFB1800400）——大规模安全可信寻址路由关键技术。参与。
- **企业合作项目**（均为**项目负责人**）：中国电信——大规模广域网感知与优化技术研究；腾讯科技——大规模微服务系统可观测性技术；深信服——应用行为感知的广域网传输优化技术；绿盟科技——安全试验设施关键技术研究。

生产系统与开源成果

- **DeepFlow**——面向云原生与 AI 基础设施的零代码 eBPF 可观测性平台（Kubernetes、服务网格、Serverless 平台、LLM / AI 智能体流水线）。**CNCF Sandbox 项目**；GitHub 4000+ stars、450+ forks；部署于数万台主机。SIGCOMM'23
- **DeepTrace**——面向智算中心与大规模微服务的无侵入分布式应用追踪与故障监控；高并发下准确率超过 95%，已被数十家企业用于生产环境。170+ stars。SIGCOMM'25、ICSE'26
- **SRmesh 探测工具包**——切片并行 SRv6 探测，以 $O(1)$ 轮次获得全网状时延图；在国家级 FITI 试验设施上验证。ICNP'25、ToN'26
- **IScan**——持续的互联网主动 / 被动探测（IP 空间、BGP/RPKI 路由、CT 日志与 Web-PKI），汇聚为可查询的互联网知识库。WWW'25、WWW'26

- **Kuiper / 智算中心智能运维**——整合故障管理、安全事件响应、配置管理与网络验证的全栈 AIOps 平台；在生产 GPU 集群环境中将 MTTR 降低 5×。NSDI'27、ICNP'25、ToN'26
- **SiteGuard / 智算中心出入口防护**——智算中心出入口流量监测与机器学习威胁检测，可运行于可编程交换机（P4/Tofino）、DPU、SmartNIC 与 eBPF，达到 Tbps 量级线速。SIGCOMM'25、SIGCOMM'26
- **NetVerify 套件**——面向配置、覆盖网与定量策略的静态验证与仿真工具族，已应用于 FITI 与 CERNET。TIFS'24、CoNEXT'25、INFOCOM'26
- **公开数据集**——GlassMiner Looking-Glass 语料（WWW'26）；带标注的加密流量 IDS 基准数据集（SIGCOMM'25/'26、CCS'21、NDSS'23）。

成果转化与国家级影响

- **中国电信骨干网**——线速加密流量异常检测；大规模广域网感知与优化。
- **未来互联网试验设施（FITI）**——可编程交换机入侵检测与分段路由诊断在国家下一代互联网试验设施上部署；配置与覆盖网验证。
- **CERNET / CERNET2**——基于 DeepFlow 的可观测性与路由异常分析，覆盖中国教育和科研计算机网。
- **腾讯云**——大规模微服务系统可观测性；跨数据中心广域网流量工程联合设计。
- **深信服**——面向企业 SD-WAN 产品的应用感知广域网传输优化。
- **绿盟科技**——安全试验设施与大规模流量分析关键技术。
- 相关技术累计服务于电信运营商、国防军队、教育网与公有云，**惠及数亿终端用户**。

发明专利

形成两个专利组合——**网络测量与安全**（27 项）与**网络智能**（19 项），其中绝大多数为第一发明人。代表性专利：

- 基于可编程交换机的高速加密流量双平面特征提取技术。
- 基于频域增强与单类自适应学习的加密流量异常检测技术。
- 基于内核态与用户态双路径协议特征解析的分布式请求追踪方法。
- 基于本地异常感知的元数据驱动型分布式请求追踪方法。
- 面向 SD-WAN 的基于 AE-DQN 的双阶段自适应异常流量检测系统。
- 深度学习恶意流量检测的预训练数据源审计方法。
- 基于 ACME 协议的安全验证方法、系统及相关装置。
- 面向跨域分段路由的可信验证方法技术。
- 基于带内网络遥测的 OSPF 路由异常检测方法技术。
- 一种面向网络快速重路由的完美路由树构建方法。
- 互联网测量服务挖掘方法；互联网路由器地理定位方法。
- 服务等级目标违规诊断方法、装置、设备及介质。

学术服务

- **程序委员会共同主席**——ACM SIGCOMM NGNO Workshop 2025。
- **差旅资助共同主席**——ACM APNet 2025。
- **程序委员会委员**——USENIX NSDI 2027（Spring）；USENIX Security 2027；IEEE INFOCOM 2023、2024、2025、2027；ACM CoNEXT 2025、2026；IEEE ICNP 2024、2025。
- **期刊编委**——《通信技术》。**客座编辑**——MDPI *Electronics*；*IEEE Journal on Selected Areas in Communications*（JSAC）。
- **期刊审稿人**——IEEE/ACM Transactions on Networking；IEEE Transactions on Parallel and Distributed Systems；IEEE Transactions on Information Forensics and Security。

教学工作

2023、2024 秋季学期	**互联网工程**——本科生课程，32 学时。涵盖路由、流量工程、可观测性与互联网测量，含实验环节。
2022–2024 秋季学期	**网络优化理论与方法**——研究生课程，32 学时。
2022–2025 春季学期	**网络异常检测**——研究生课程，32 学时。

指导学生

在读（13 人）

赵铠阳	博士 · 五年级	SRv6 探测（ICNP'25、ToN'26）；与尹霞教授、吴建平院士联合指导
耿彦涛	博士 · 四年级	DeepTrace 负责人（SIGCOMM'25、ICSE'26）
高杨宸严	博士 · 四年级	与赵有健教授联合指导
刘雪枫	博士 · 三年级	网络验证
肖泽基	博士 · 三年级	与王继龙教授联合指导
Boyang Wang	博士 · 二年级	与尹霞教授联合指导
童彦图	博士 · 一年级	SRv6 / 网络测量
李 卓	硕士 · 三年级	加密流量入侵检测
Hanyang Lin	硕士 · 二年级	eBPF 可观测性
武志衡	硕士 · 二年级	分布式追踪
钱林强	硕士 · 一年级	可编程数据平面
佟 瑶	硕士 · 一年级	与尹霞教授联合指导
彭可心	硕士 · 一年级	与赵有健教授联合指导

已毕业（6 人）

沈俊贤	博士	华南理工大学助理教授 （SIGCOMM'23、'25；ToN'26）
Heng Yu	博士	中关村实验室 （TPDS'23）
Yi Qiao	博士	中国航天科技集团 （INFOCOM'24；TPDS'22）
Shangbin Han	博士	泉城实验室 （TIFS'21）
汤之韫	硕士	华为技术有限公司
李嘉睿	硕士	阿里云

正在招收 2027 年秋季入学的博士 / 硕士研究生，具备系统编程经验者尤佳。